Crowdsourced Mosquito Sightings as an Early Signal of West Nile Virus

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1 Introduction

In the summer of 1999, West Nile virus (WNV) was found in the United States for the first time, in New York City. That first year, CDC records show 62 human cases. Within four years the virus had spread across the country, and in 2003 alone the CDC recorded 9,862 cases. [1] It has stayed ever since. Between 2016 and 2025 alone, ArboNET, the national arboviral surveillance system run by the CDC and state health departments, recorded 19,232 human cases, 13,063 of which were neuroinvasive. [1]

Around a decade later and on the other side of the country, a group of graduate students at UC Berkeley launched a master's final project called iNaturalist in 2008. [2] The idea was simple: anyone could photograph a plant or animal, upload it with a time and a place, and let a community of naturalists help identify it. The project has since grown into one of the world's largest biodiversity platforms, and its users now photograph mosquitoes along with everything else. None of these users are trying to do disease surveillance. They are documenting what they see.

This brings us to our research question: can the mosquito sightings that iNaturalist users post for biodiversity reasons act as an early signal of West Nile virus activity? More specifically, does an unusually mosquito-heavy spring on iNaturalist precede a worse WNV season in the same place? The hypothesis that we had and intended to test was that it would, and that the signal would be strongest for mosquitoes of the genus *Culex*, which are the main carriers of the virus.

In doing so, we find that the answer is mostly no. Mosquito-heavy seasons on iNaturalist do tend to be worse WNV seasons nationally, but the crowd signal does not tell you which states or counties are in for a bad year, and it adds no forecasting skill beyond a state's own history.

2 Background

West Nile virus cycles between birds and mosquitoes, especially *Culex* species. Some infected birds develop high levels of virus in their blood, mosquitoes that bite them become infected, and those mosquitoes can then pass the virus on to more birds and eventually to people. [3] This is what made the idea plausible to us in the first place: if more mosquitoes are around and noticeable, more people might photograph them, and more transmission might follow.

The idea of finding a disease signal in online behavior is not new. Ginsberg et al. showed that Google search queries could track influenza-like illness one to two weeks ahead of CDC reports. [4]

Goel et al. later argued that a signal like this is only useful if it beats a simple baseline, and found that a plain autoregressive model did almost as well as search data for flu.[5] We took both lessons into our design. Our crowd signal comes from a source that was never meant for epidemiology, and we judged it against the simplest baseline we could think of, which is what each state's WNV level usually is.

3 Online Data and Resources

In order to begin collection of our data, we invoked the iNaturalist API[6] through a series of python scripts. We pulled every verifiable US observation of the mosquito family (Culicidae) from 2015 through 2025, which came out to 76,129 observations posted by 23,144 different people. For each observation we kept the date, an anonymous observer ID, the taxon it was identified as, its quality grade, and its coordinates. Since iNaturalist asks for no more than about one request per second, we split the pull into four date ranges and ran them side by side.

A count of mosquito photos on its own does not mean much without knowing how many people were out photographing insects in general, so we also pulled denominators from the API's histogram endpoint. These were monthly counts per state of all insect observations (about 30 million in total), of crane-fly observations, and of all observations, as well as yearly insect counts for each of the 2,357 counties where at least one mosquito had been recorded.

For West Nile virus, we used the CDC's ArboNET historic data,[1] which was finalized on September 16, 2026. The dashboard on that page is built on downloadable files that give human cases, neuroinvasive cases and non-human activity for every county and year from 1999 to 2025, as well as national case counts by month of illness onset. We chose neuroinvasive cases as our outcome, because the CDC itself notes that milder, non-neuroinvasive cases are under-reported unevenly and should not be compared across places or over time.[1] Finally, we used 2016 Census county population estimates[7] to turn county case counts into incidence rates.

To link the two sources, we placed each mosquito observation in a county by matching its coordinates against Census county boundaries, which placed 99.5% of them. For the 11.8% of observations whose exact location iNaturalist hides for privacy, we used iNaturalist's own county assignment instead. As a validation step, our state totals came within 0.3% of iNaturalist's own counts, and our county case totals added up exactly to the national totals the CDC publishes (for example, 2,167 cases and 1,517 neuroinvasive cases in 2025).

4 Our Original Question and How It Changed

Our proposal asked whether spatiotemporal sightings of *Culex* mosquitoes could predict WNV cases at the level of a county-week. We planned to compare several crowd measures, namely raw observation counts, the number of unique observers, changes from a location's normal level, and mosquito observations as a share of local iNaturalist activity. Once we actually had the data, three problems forced us to change the question.

1. ***Culex* is too rare a label.** Only 10.0% of mosquito observations were identified as *Culex* (7,585 over ten years). Most of what people photograph is *Aedes* (39.6%) or *Psorophora* (11.9%), and another 11.7% are never identified past the family level. We therefore made all mosquitoes our main signal and kept *Culex* as a sub-signal.

2. **County-weeks are almost always empty.** The median county has only 6 mosquito observations in ten years, and only 2.6% of county-weeks have any at all. Even Cook County, Illinois, home to Chicago, has 479. On top of this, the CDC's public data only goes down to county-year, and the timing of illness onset is only published nationally by month. We moved to state-years (50 states and DC over ten years), national months, and a pooled comparison across counties.

3. **Raw counts measure the app, not the mosquitoes.** Mosquito observations grew from 1,156 in 2016 to 16,770 in 2025, roughly fourteen-fold, while WNV cases showed no trend at all (Figure 1). To deal with this, we divided each mosquito count by the number of insect observations from the same place and months, which we call the mosquito share, and we compared every state against its own normal level and against the national year.

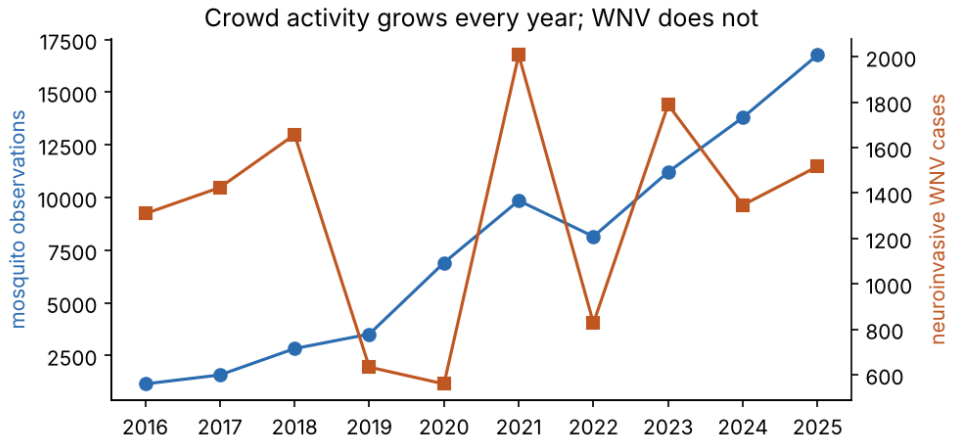


Figure 1: Mosquito observations on iNaturalist per year (blue) against neuroinvasive WNV cases reported to the CDC (orange), 2016–2025. Crowd activity grows every year while WNV moves up and down.

The question we ended up asking was this: within a state, does an April–July mosquito share that is higher than usual, in a year that is not unusual nationally, go along with more neuroinvasive WNV cases that year? We picked April through July because human cases peak in August and September, so a signal from those months would have to come first. Alongside this main test, we also asked whether mosquito sightings lead cases over the course of a season, whether mosquito-heavy years are worse WNV years nationally, and whether counties with a higher mosquito share see more WNV. We kept the proposal's other crowd measures as alternatives, and we added crane flies, a common mosquito look-alike that carries no disease, as a placebo.

5 Methods

Our mosquito share was computed as (mosquito observations + 0.5) divided by (insect observations + 1), so that a state with no mosquito photos in a given window would not produce a zero, and we analyzed it on a logarithmic scale. The outcome was log(1 + neuroinvasive cases). For the main test, we fit a regression of the outcome on each April–July crowd signal with fixed effects for each state and each year, and clustered the standard errors by state. Importantly, the state effects absorb how WNV-prone a state normally is and how popular iNaturalist is there, while the year effects absorb nationally good and bad seasons. We only kept state-years with at least 200 insect observations in April–July, which removed 2 of the 510.

We also ran a number of robustness checks. These included restricting to 2020–2025 when iNaturalist volume was large, a negative binomial model on the raw case counts, a version without year effects, and within-state rank correlations for each state. Following the lesson from Goel et al., we ran a leave-one-year-out forecast in which each state's season was predicted either from that state's average in the other years alone, or from that average plus the mosquito anomaly, and compared the errors.

At the national level, we built monthly anomalies for May through October by subtracting each calendar month's ten-year average from both the mosquito share and the case counts. Since ten seasons is not a lot of data, we tested these correlations with a permutation test that shuffles which year's mosquito series is paired with which year's cases. For counties, we pooled 2016–2025 and compared the mosquito share with average annual incidence across 2,160 counties that had at least 300 insect observations, controlling for population, overall insect activity and state. We left out Connecticut because its county codes changed in 2022.

6 Results

The results we found mostly did not support our hypothesis. Every mosquito signal pointed in the direction we expected, but none of them held up once we accounted for what a state's WNV level usually is.

6.1 Seasonal timing

Mosquito observations center on mid-July, about 1.1 months before the center of neuroinvasive onsets in mid-August (Figure 2a). At first glance this looks like a lead, but insect observations in general center on early July, so most of this lead is simply when people are outside taking pictures of bugs. Mosquito observations also stay roughly flat from June through September rather than climbing toward the WNV peak. Surprisingly, *Culex* observations peak in October, when 16.5% of them were posted, which is after the August peak that holds 38.9% of onsets. One possible explanation is that *Culex* females come indoors in the fall to wait out the winter, where people are more likely to notice them.

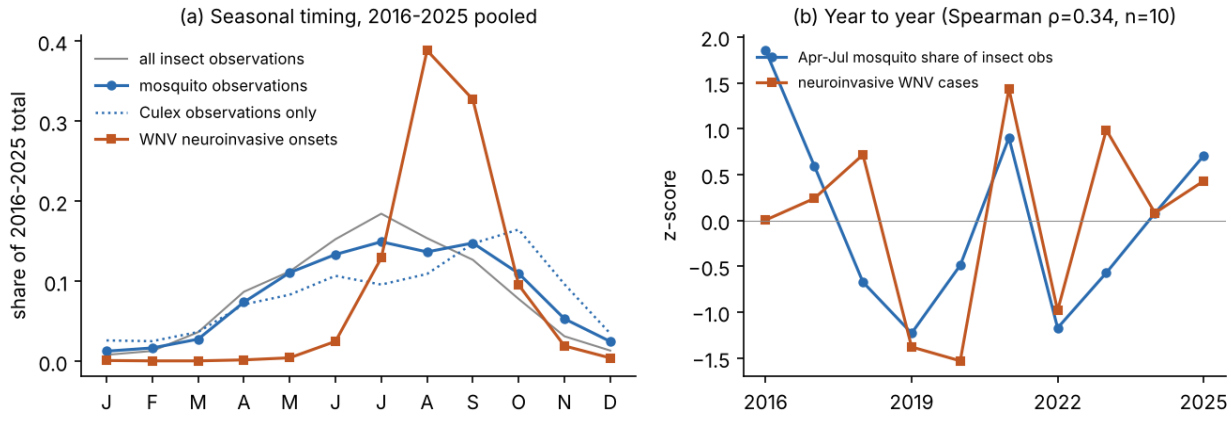


Figure 2: (a) Share of all 2016–2025 records falling in each calendar month for insects, mosquitoes, *Culex* and neuroinvasive WNV onsets. (b) Year-to-year z-scores of the national April–July mosquito share and neuroinvasive cases.

6.2 Year to year nationally

At the national level, there is a real but loose link. The April–July mosquito share and that year's neuroinvasive cases had a rank correlation of 0.35 ($p = 0.33$, $n = 10$), and the July–September share did better at 0.58 ($p = 0.08$) (Figure 2b). With the monthly anomalies, a mosquito share above its usual level was followed by more cases one and two months later ($r = 0.39$ and 0.44), and this survived the permutation test ($p = 0.02$ for both). However, when we also removed each year's average, the correlation dropped to between 0.05 and 0.15 and was no longer significant. This told us that the link is about whole seasons being mosquito-heavy, and not about month-to-month timing within a season.

6.3 State-year panel

Table 1 shows the main test. Every mosquito signal has a positive coefficient and the crane-fly placebo has a negative one, but none of them are significant, and none explain even 1% of the variation that is left after state and year effects. Counting unique observers instead of observations did not help, and neither did narrowing down to *Culex*. Figure 3 shows how little the mosquito signal stands out from the placebo.

April–July signal	Coefficient	SE	p	Within-R^2
Mosquito share of insect observations	0.072	0.063	0.25	0.003
Mosquito observers per insect observation	0.058	0.089	0.51	0.001
Culex share of insect observations	0.050	0.038	0.19	0.003
Raw mosquito observations (log)	0.090	0.073	0.22	0.005
Unique mosquito observers (log)	0.098	0.106	0.35	0.003
Placebo: crane-fly share	−0.058	0.086	0.50	0.001
Mosquito share, full year (concurrent)	−0.005	0.065	0.93	0.000

Table 1: Regressions of log(1 + neuroinvasive cases) on each crowd signal with state and year fixed effects, 508 state-years, standard errors clustered by state. Within-R^2 is the share of variation explained beyond the state and year effects.

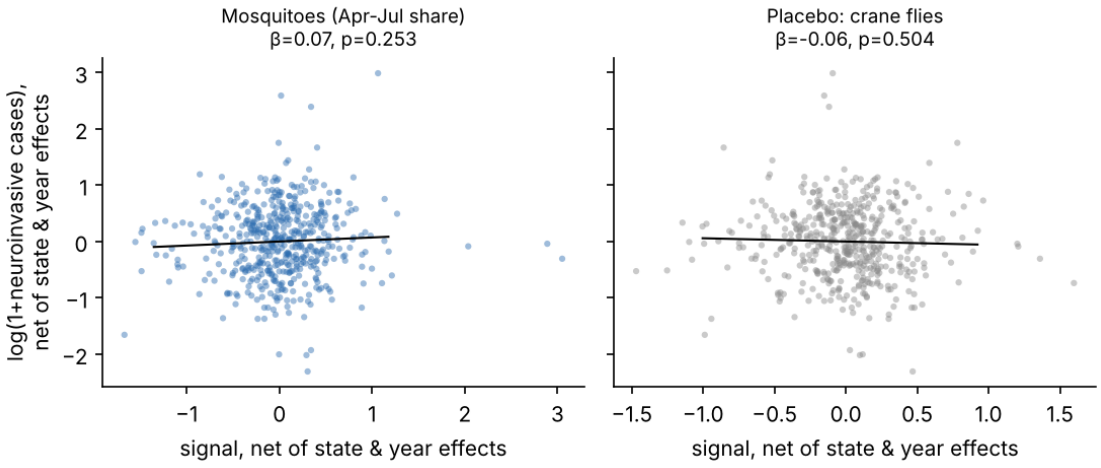


Figure 3: April–July signal against neuroinvasive cases, both net of state and year effects, for mosquitoes (left) and the crane-fly placebo (right). Each point is one state-year.

The robustness checks told the same story. Restricting to 2020–2025, when iNaturalist had enough

users for state shares to be stable, gave the strongest result at 0.25, but it still was not significant ($p = 0.08$, 306 state-years). The negative binomial model gave a coefficient of 0.20, which looks significant with the model's own standard errors ($p = 0.01$) but not once they are clustered by state ($p = 0.10$). Removing the year effects made the signal significant (0.13, $p = 0.03$), which matches what we saw nationally: the signal appears once nationally bad years are allowed to count. State by state, 28 of 46 states had a positive within-state correlation, which is not much better than a coin flip (sign test $p = 0.18$). Most telling of all, adding the mosquito anomaly to a state's own average made the leave-one-year-out forecasts slightly worse rather than better, with a mean absolute error of 0.687 compared to 0.682.

We also ran into the trap that the raw counts set. Pooled across all state-years, raw mosquito counts correlate with cases at 0.23 ($p < 0.001$), which could easily be mistaken for a finding. Once the counts are turned into a share, this drops to -0.04, because large states simply have more of both.

6.4 Counties

Counties with a higher mosquito share do have slightly higher average neuroinvasive incidence (rank correlation 0.066, $p = 0.002$) and more years with any WNV activity (0.075, $p < 0.001$). However, once we compared counties only within their own state and controlled for population and insect activity, the effect vanished (coefficient 0.006, $p = 0.57$). The raw correlation reflects which states are prone to WNV, not anything happening locally (Figure 4).

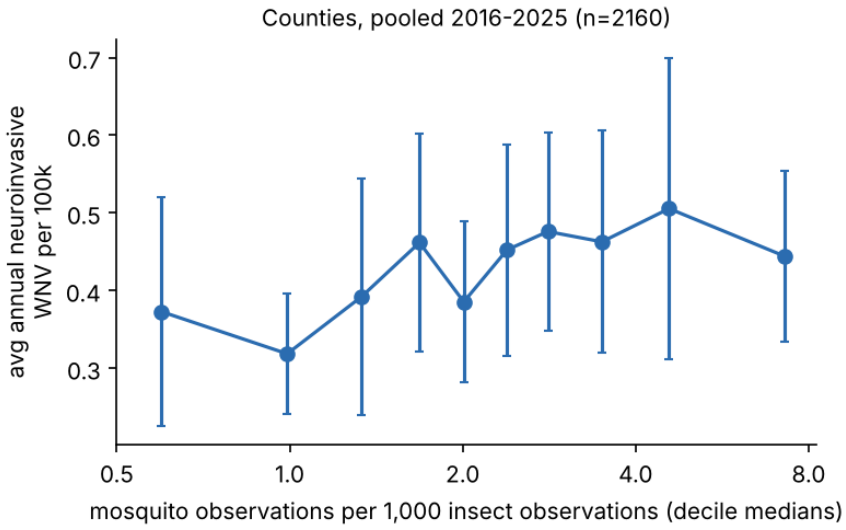


Figure 4: Average annual neuroinvasive WNV incidence per 100,000 people by decile of a county's mosquito share, pooled 2016–2025, with 95% intervals.

7 Discussion

As mentioned earlier in our paper, the crowd signal we found is real but weak, and it lives at the national level. One possible explanation for this is what people actually photograph. The observations are dominated by *Aedes* and *Psorophora*, and the *Culex* mosquitoes that carry the virus make up only a tenth of them and are posted most often in the fall, after the outbreak has already peaked. In other words, the crowd is documenting the mosquitoes that bother them, not

the ones that matter for WNV.

It is also worth noting that the crowd itself was not the problem. The 75,798 mosquito observations from 2016–2025 came from 23,078 people, 69.4% of whom posted only one mosquito, and the top ten users accounted for just 5.5% of observations. This is a genuine crowd and not a handful of enthusiasts, and counting observers instead of observations made no difference.

These findings may have broader implications for using citizen science as a public health signal. A platform like iNaturalist can tell us when and roughly where people are noticing mosquitoes, and in a broad sense that does line up with bad WNV years. However, the step from "people noticed more mosquitoes" to "more people got sick here" requires the right species, at the right time, in the right place, and our data was thin on all three. Like Goel et al. found with search data, the signal only looks impressive until it is compared to a simple baseline.[5]

However, something of note that can be expanded on with further analysis is the national pattern. A hot summer could plausibly produce both more noticeable mosquitoes and more WNV transmission at once, which would explain why the link shows up across whole seasons but not within states. Testing this would require weather data, which was outside the scope of this project. A more targeted design, such as focusing only on research-grade *Culex* observations or using weekly case data that some state health departments publish, might also find a sharper signal.

8 Limitations and Conclusion

8.1 Limitations

The biggest limitation of our study is power, since ten years means only ten national seasons. In 2016 the entire country logged 535 mosquito observations between April and July, so state shares before about 2019 are mostly noise, which may be why the 2020–2025 estimate was larger. The CDC only publishes onset timing nationally and only yearly totals by state and county, so a true lead test at the state-week level was not possible with public data. Cases are also counted by county of residence rather than where the person was bitten, which blurs the county comparison. On the iNaturalist side, 60% of the observations have not reached research grade, meaning the community has not yet agreed on their identification, so genus labels are imperfect. Finally, our county boundaries were generalized, which may misplace a small number of observations near county lines, and our county populations are from a single year.

8.2 Conclusion

iNaturalist users posted over 76,000 mosquito observations in the US without ever trying to track a disease, and in a broad sense their activity does line up with West Nile virus: mosquito-heavy seasons nationally tend to be worse WNV seasons. However, our hypothesis did not hold where it mattered. Within a state, an unusually mosquito-heavy spring did not predict a worse season, *Culex* sightings were too rare and came too late, and adding the crowd signal made forecasts slightly worse than simply using a state's history. That suggested to us that the deciding factor is less the size of the crowd and more so whether the crowd is looking at the right thing.

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A Effort

Most of our time went into getting the data and making sure the two sources actually lined up. The iNaturalist API limits requests to about one per second, so pulling 76,129 observations, monthly histograms for every state, and yearly insect counts for 2,357 counties took close to an hour of collection even with four parts running side by side, and we built the collector so that it could resume after dropped connections. The CDC's county data lives inside an interactive dashboard, so we had to find the data files that sit behind it on the historic data page. We then validated both sources against their published totals before trusting anything.

Besides that, we spent a long time on things that the final analysis does not show. Our proposed design, *Culex* sightings at the county-week level, did not work because *Culex* made up only a tenth of the observations and almost every county-week was empty. Our first comparisons of raw counts against cases looked like a strong relationship, which turned out to be state size and iNaturalist's growth. Switching to the mosquito share with state and year effects, and adding the crane-fly placebo and the out-of-sample forecast, was the project's main turning point.

B Use of AI and Other Online Resources

We used Claude[8] to help write the data collection and analysis scripts, run the statistical analysis, and draft this report and our summary slide from our proposal and results.

Other resources: the iNaturalist API[6], the CDC West Nile virus historic data[1], documentation for pandas[9], statsmodels[10], shapely[11] and matplotlib[12], and the papers we read in class, especially Ginsberg et al. (2009)[4] and Goel et al. (2010)[5].